COMP7505 Group Project Report

InterviewAI — Your AI Interview Prep Coach

Track A: Smart Adaptive Interfaces | Group: Cre8

Course: COMP7505 User Interface Design and Development

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Deliverable 1: User Needs & Project Definition

This deliverable documents the user-research foundation for InterviewAI. It defines the primary target user, summarises the main pain points identified through the questionnaire and interviews, specifies the core tasks the system is intended to support, and clarifies the scope and boundaries of the proposed design. The findings presented here provide the basis for the interaction design, evaluation, and roadmap discussed in Deliverables 2-5.

1.1 Target User Persona

The primary persona, Kevin Li, was developed from a pre-project questionnaire completed by nine respondents and from follow-up semi-structured interviews. All respondents were fresh graduates or near-graduates from computing-related backgrounds, and most were master's students at HKU. The persona consolidates the recurring goals, behaviours, and concerns observed across these data sources.

Attribute	Detail
Name	Kevin Li
Age	23
Occupation	Fresh graduate in Computer Science
Location	Hong Kong
Job-seeking Status	Actively interviewing for front-end / full-stack developer roles
Core Goals	Kevin wants to prepare for both common and role-specific interview questions before formal interviews. He wants to improve the clarity, structure, and naturalness of his answers through repeated practice. He also expects the system to provide personalized and trustworthy feedback grounded in his resume and the job description, while helping him track progress across multiple sessions.

Pain Points

#	Pain Point	Severity
1	Existing AI feedback is often generic and does not explain what specifically should be improved.	High
2	Current tools do not sufficiently incorporate the user's resume, project experience, or job-description context when generating questions or feedback.	High
3	AI-generated sample answers often sound formulaic and do not reflect the user's personal background or communication style.	Medium

4	Users often find it difficult to judge whether AI-generated feedback is reliable or how confident the system is in its evaluation.	Medium
5	Mock interview tools often fail to reproduce realistic follow-up questioning and the level of pressure found in actual interviews.	Low

Representative User Quote: "We do not only want AI-generated answers; we also want interview feedback that feels personalized, realistic, and trustworthy."

1.2 Core User Tasks

The core end-to-end task is to prepare for a specific interview with AI support. In practice, users first provide the target role, company, job description, and relevant resume or background information, and then continue through one of the two workflows below.

Task A: Practice

In the practice workflow, the user reviews AI-generated interview questions tailored to the target role, answers them through text or voice, and receives scores, explanations, confidence cues, and suggested revisions. The system can then provide more detailed feedback or optimized answer suggestions, and the session can be saved for later review.

Success Criteria: The user completes at least one full question-answer-feedback cycle and receives feedback that is specific enough to guide the next revision.

Task B: Review

In the review workflow, the user uploads a mock interview recording, after which the system extracts the interview questions and the user's answers for review. The user can then check and edit the extracted content before receiving feedback, polishing suggestions, and a structured review summary.

Success Criteria: The user uploads an audio file, verifies the accuracy of the extracted interview content, and receives a structured performance summary highlighting both strengths and areas for improvement.

1.3 Project Scope & Boundaries

Dimension	Boundary
Usage Scenario	The proposed design supports interview preparation before formal interviews and post-interview reflection, rather than use during a live interview.

Device	The current concept is designed primarily for laptop-based preparation, while phone use is treated as a secondary scenario for quick revision or last-minute checking.
Technical Constraints	The prototype assumes stable internet access and allows certain AI capabilities to be simulated; a production-ready AI backend is outside the scope of the current deliverable.
Design Assumptions	AI is positioned as a support tool rather than a final decision-maker, so the system should explicitly indicate when its confidence is low or when user context is insufficient.
Excluded Requirements	The project does not aim to become a full recruitment platform, a verified database of company-specific interview questions, or a real-time video interview system.

Deliverable 2: Interaction Design Specification & Prototype

This deliverable translates the D1 research foundation into a concrete interaction design solution. D1 identified Kevin Li as a laptop-first fresh graduate who needs JD/resume-aware practice, trustworthy feedback, user control over AI judgement, and recovery when voice or AI functions fail. Therefore, this section documents the interaction model, information architecture, navigation structure, complete Practice and Review workflows, and the high-fidelity prototype coverage directly in the written report, rather than relying on content outside the submitted prototype evidence. It follows the standalone D2 interaction-design document: Design Spec, Flow Diagrams, and Prototype evidence are treated as one traceable package, so every major D2 design decision can be mapped back to D1 user needs and to the course requirements for page jumps, operational feedback, and exception branches.

D1-to-D2 Traceability

The D2 design is not a separate feature list. Questionnaire and interview findings showed that candidates worry about unreliable AI feedback, generic suggestions, weak use of resume/JD context, and unrealistic practice pressure. The D2 response is to make AI reasoning visible, keep confidence cues next to scores, define capability boundaries in plain language, and provide user controls such as flagging, overriding, undoing, and discarding AI outputs.

The primary persona is a laptop-first fresh graduate preparing iteratively before interviews. This directly motivates the top-level IA: Dashboard for entry, Practice for JD/resume-aware preparation, Review for recorded-interview reflection, History for repeated improvement, and Settings for confidence and confirmation preferences.

The two D1 core paths are preserved in D2 terminology. Practice means role/JD/resume context to tailored questions, answer input, feedback, user control, and History. Review means recording upload, transcript extraction, editable Q&A, structured summary, and export or follow-up practice. Flowchart labels and prototype screens use the same names so reviewers can trace each screen back to the D1 task and success criteria.

2.1 Interaction Model

InterviewAI uses a human-AI collaborative model. The AI acts as a preparation coach that proposes questions, scores, explanations, and summaries, while the user remains the final judge of whether the output is useful or correct. This model directly responds to D1 pain points: users doubted black-box scores, wanted feedback grounded in JD/resume context, and wanted the ability to challenge wrong AI advice. It also establishes the design logic later formalised in D4: visible AI reasoning, preserved user control, progressive disclosure, and graceful degradation.

Transparency — Every important AI output is paired with a short rationale and a confidence cue. This prevents the system from behaving like an unexplained scoring engine and helps users decide whether to trust or verify a suggestion.

User Control — Users can flag inaccurate feedback, override scores, undo actions, discard evaluations, edit extracted content, and move from History back into targeted practice. AI suggests; the user decides.

Graceful Degradation — When voice recognition, question generation, extraction, or network-dependent AI feedback fails, the interface preserves user progress and offers recovery actions such as retry, text fallback, a general question bank, or manual Q&A entry.

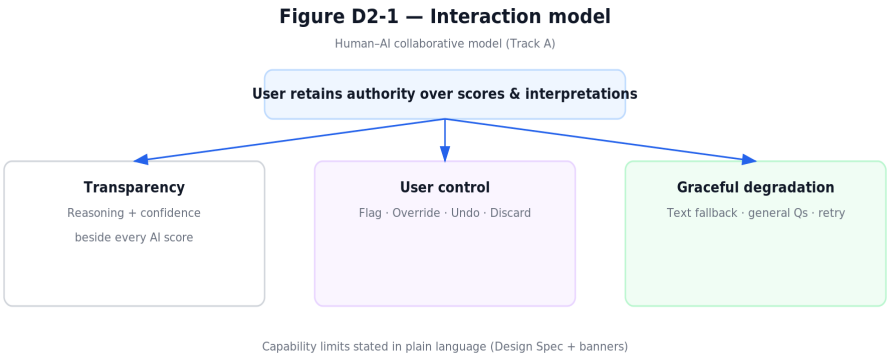

Figure D2-1: Interaction Model — Transparency, User Control, Graceful Degradation

2.2 Information Architecture

The top-level IA contains Dashboard, Practice, Review, History, and Settings. This is not feature stacking: each area maps to a user need from D1. Dashboard provides a low-friction entry point; Practice supports the primary JD/resume-to-feedback journey; Review supports the secondary recorded-interview journey; History closes the iterative learning loop; Settings exposes AI confidence and confirmation preferences. Exception states are part of the IA itself, including invalid files, AI generation error, voice failure, low audio quality, and extraction failure.

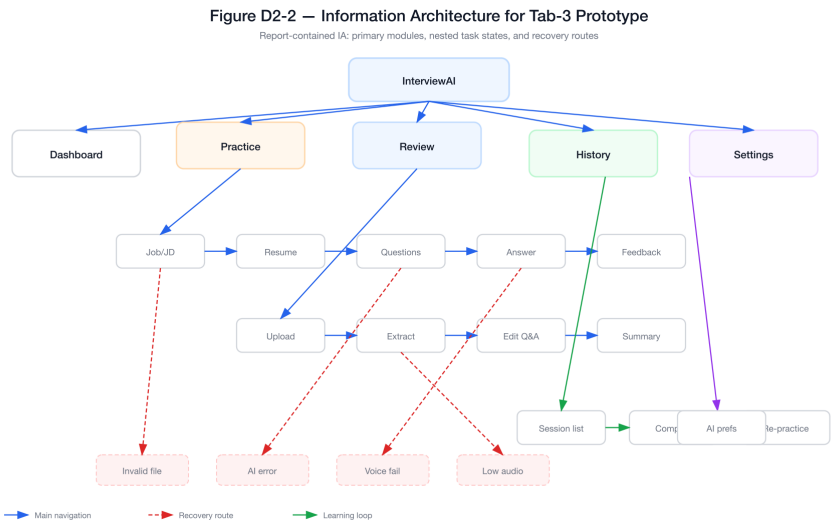

Figure D2-2: Information Architecture for the Submitted Prototype

2.3 Navigation Structure

Navigation is structured around one persistent sidebar and two guided linear workflows. Practice uses Job/JD Input -> Resume Upload -> Question List -> Answer Page -> Feedback, while Review uses Upload Recording -> Extracted Q&A -> Editable Transcript -> Review Summary. Multi-step flows use breadcrumbs and Next / Back controls; global navigation remains stable so the user can recover orientation after switching modules. Each page jump has a trigger and operational feedback: inline validation, loading states, confidence warnings, autosave, undo toast, or unsaved-draft confirmation. Figure D2-3 records from-state, to-state, user trigger, and transition feedback directly in the report.

Figure D2-3 — Navigation Structure and Page-Jump Matrix

Both D1 core workflows are documented with trigger and operational feedback

From	To	User trigger	Operational feedback
Dashboard	Job/JD input	Start Practice	Slide; breadcrumb Step 1
Job/JD input	Resume upload	Next after required fields	Inline validation; Next enabled
Resume upload	Question list	Generate questions	Loading; context summary visible
Question list	Answer page	Select question	Question status set In progress
Answer page	Feedback page	Submit answer	Evaluation loading; draft saved
Feedback page	Question list / History	Next / Save	Score saved; undo toast
Dashboard	Upload recording	Start Review	Slide; upload rules shown
Upload recording	Extracted Q&A	Upload valid file	Progress indicator; audio check
Extracted Q&A	Review summary	Confirm edited transcript	Confidence warning if audio low
Review summary	History / Practice	Save / Practice weak areas	Summary saved; weak-area path opened
Any task page	History / Settings	Sidebar click	Warn if unsaved draft exists

Navigation evidence is in the report, not dependent on tabs 1 or 2

Figure D2-3: Navigation Structure and Page-Jump Matrix

2.4 Core Task Workflows

Practice Workflow

The Practice workflow is the main D1 core task: the user enters role and JD context, uploads or confirms resume context, generates tailored questions, answers by text or voice, receives AI evaluation, reviews confidence and reasoning, applies user-control actions, and either continues to the next question or saves the session to History. The workflow includes page jumps, operational feedback, and exception branches: missing JD fields produce inline errors; invalid resume files show accepted formats; AI generation errors offer retry or a general question bank; voice failure preserves partial transcript and switches to text; low-confidence feedback displays a limitation banner and invites user verification.

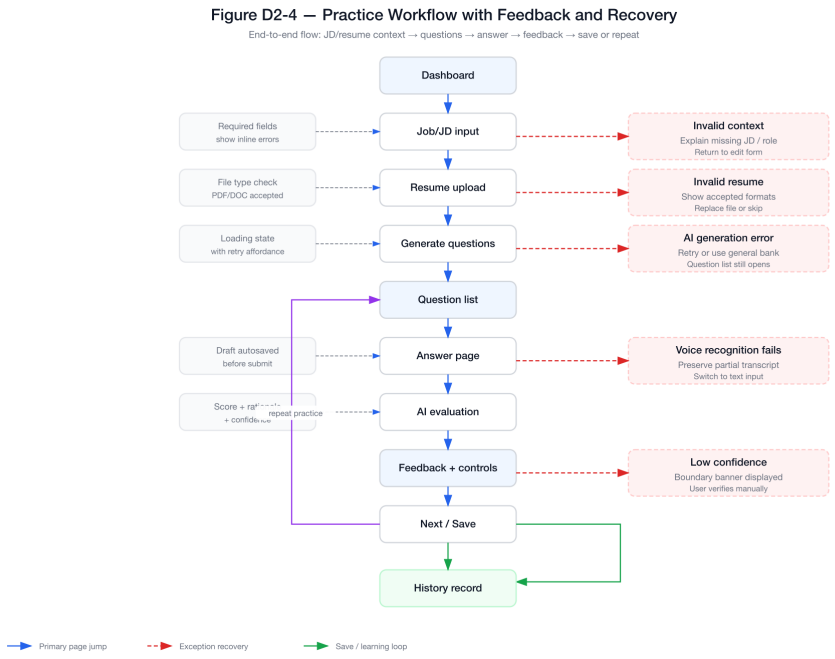

Figure D2-4: Practice Workflow with Feedback and Recovery

Review Workflow

The Review workflow supports the recorded-interview use case from D1. The user uploads a recording, the system validates file type and audio quality, extracts Q&A pairs, allows the user to edit the transcript, generates a review summary, and saves or exports the result. Exception branches are explicitly recoverable: invalid files can be replaced; low audio quality is shown before the user trusts the output; extraction failure opens manual Q&A entry; AI summary failure preserves the edited transcript and allows retry. This keeps the Review path aligned with the D4 principle of graceful degradation.

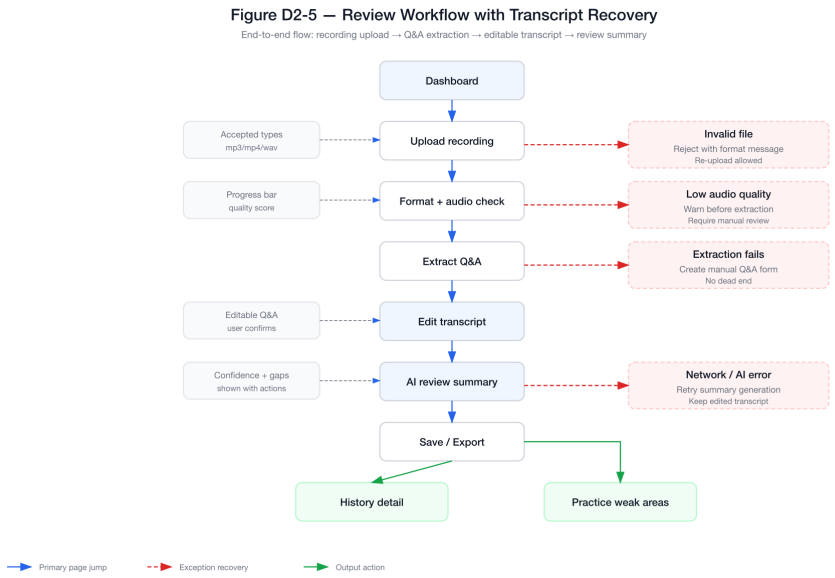

Figure D2-5: Review Workflow with Transcript Recovery

2.5 Interactive Prototype

The submitted interactive artefact is the high-fidelity Prototype experience in index.html, Tab 3: Prototype. The file is designed as a self-contained HTML prototype with simulated AI and error paths, so no backend is required for assessment. It covers Dashboard, the full Practice path, the Review path, History, Settings, simulated failure modals, confidence indicators, and user-control actions. Because the prototype tab is treated as the main interactive submission evidence, the interaction specification, IA, navigation matrix, workflow diagrams, and Track A rationale are also presented directly in this report section so the written portfolio remains self-contained.

Prototype submission note: open index.html locally, select Tab 3: Prototype, and use the visible controls to demonstrate the primary Practice flow, the Review flow, and selected failure states such as voice-recognition failure and AI-generation error. If the local HTML prototype cannot be opened, use the online backup link: https://ui-mocha-seven.vercel.app.

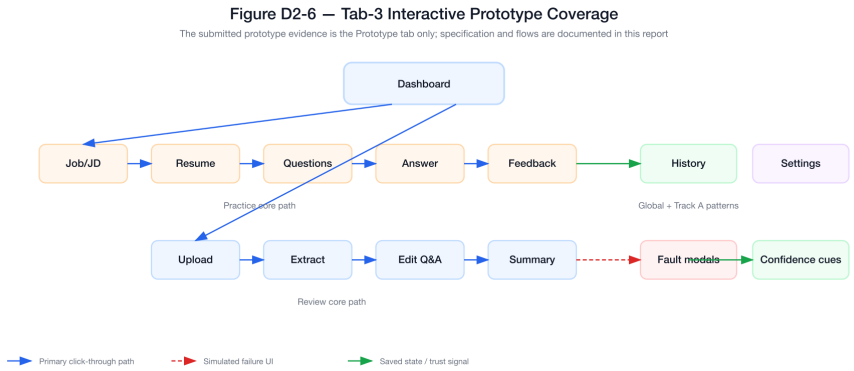

Figure D2-6: Tab-3 Interactive Prototype Coverage

2.6 Track A Compliance

Track A Requirement	Implementation
Capability Boundaries	Can: assess observable communication qualities such as answer structure, logical flow, clarity, STAR usage, and relevance to the user-provided JD/resume context. Can: generate role-relevant interview questions from the information supplied by the user. Cannot: verify deep technical expertise, infer a real interviewer's hidden intent, replace human judgement, or guarantee interview outcomes. Limited: voice and transcript analysis depend on audio quality, accent, noise, and recording conditions; low-confidence results therefore require user verification.
User Control	Every AI output supports user agency: Flag Inaccurate, Override Scores, Undo, Discard, edit extracted Q&A, confirm sensitive actions, and use History to restart targeted practice.

≥ 2 System Failure Solutions	Voice recognition failure → preserve partial transcript and switch to text input. AI generation error → retry or use a general question bank. Extraction failure → manual Q&A entry. Invalid upload → explain accepted formats and allow re-upload.
≥ 1 Limitation Mechanism	Confidence indicators, low-confidence copy, audio-quality warning, and capability-boundary banners tell users when AI output should be treated as uncertain or externally verified.

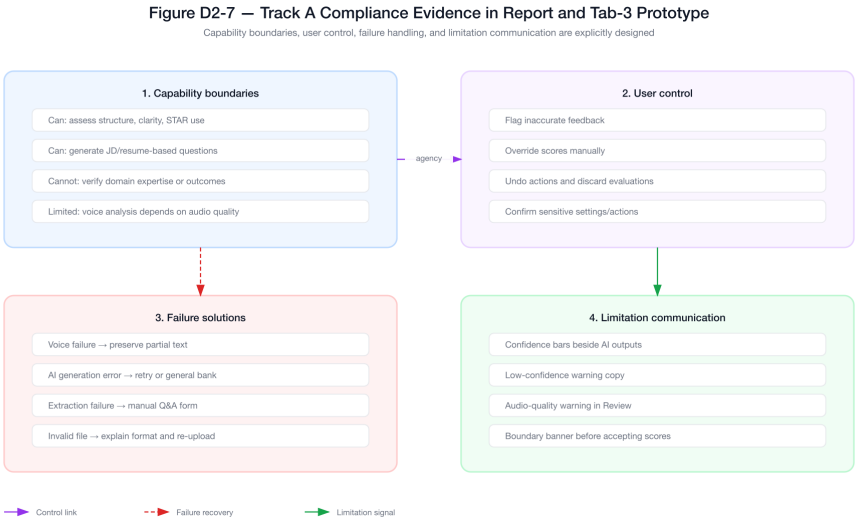

Figure D2-7: Track A Compliance Evidence — Failure Handling and Limitation Communication

2.7 Alignment with Interaction Design & Prototype Quality

This D2 section directly addresses the grading focus for Interaction Design & Prototype Quality. Interaction logic is supported by linear guided flows, explicit confirm/retry/fallback behavior, and AI outputs tied to confidence and user actions. Information architecture is clarified through one top-level IA with nested Practice and Review flows. Prototype completeness is demonstrated through both core journeys, History, Settings, and selected failure UIs. Workflow integrity is maintained because the flowcharts, navigation matrix, and prototype screens describe the same Practice and Review paths.

Figure evidence summary: D2-1 explains the interaction model; D2-2 shows information architecture and exception paths; D2-3 records navigation transitions and feedback; D2-4 and D2-5 document the Practice and Review workflows; D2-6 shows prototype coverage; D2-7 connects Track A failure handling and limitation communication to concrete interface evidence.

Deliverable 3: Design Evaluation & Iteration

This deliverable documents how the InterviewAI prototype was evaluated, what usability problems were discovered, and how those findings were translated into concrete design iterations. The evaluation focused on whether first-time users could understand the AI-supported workflow, maintain control over their interview context and answers, recover from errors, and receive feedback that was specific enough to support improvement. To avoid relying on a single source of evidence, we combined expert inspection, task-based walkthrough, and participant-based observation.

The final iteration direction was guided by one central design question: how can InterviewAI behave less like a black-box scoring tool and more like a transparent, controllable interview coach?

3.1 Evaluation Plan

The evaluation plan was designed to test the two core workflows defined in Deliverable 2: Practice, where users prepare with AI-generated interview questions and feedback, and Review, where users upload or review a mock interview and receive structured performance insights. The evaluation also checked the Track A requirements around capability boundaries, user control, failure handling, and limitation communication.

Evaluation Objectives

- Assess learnability: whether a new user can start a practice session, answer a question, and understand the feedback without external instruction.
- Assess trust and transparency: whether AI scores, suggestions, and confidence indicators provide enough reasoning for users to judge reliability.
- Assess user control: whether users can edit context, preserve drafts, challenge AI outputs, and recover from mistaken actions.
- Assess cognitive load: whether feedback, coaching panels, and history views are informative without overwhelming the user.
- Assess failure handling: whether recognition or generation failures lead to useful fallback actions rather than dead ends.

Method	Participants / Evaluators	Focus	Evidence Collected
Heuristic Evaluation	3 evaluators from the project team	Expert inspection using Nielsen's 10 heuristics, summarized into six usability dimensions.	Severity ratings, violated heuristic, screen location, and recommended fix.
Cognitive Walkthrough	2 evaluators acting as first-time users	Step-by-step learnability of the Practice and Review workflows.	Predicted user intention, visible action, feedback clarity, and likely breakdown point.
Think-Aloud Protocol	3 participants matching the D1 persona	Observed use with realistic interview-preparation tasks.	Verbal comments, hesitation points, errors, recovery attempts, and post-task preference notes.

Task Scenarios

- Scenario A - Practice: The participant prepares for a front-end or full-stack developer interview, enters job and resume context, generates questions, answers one question, and reviews AI feedback.
- Scenario B - Review: The participant checks an uploaded mock-interview transcript, verifies the extracted question-answer pairs, and reviews the system's summary of strengths and improvement areas.
- Scenario C - Recovery: The participant encounters a simulated recognition or generation issue and chooses an appropriate fallback path.

Procedure and Metrics

Each session followed the same sequence: briefing, task execution, observation, short debrief, and consolidation of findings. We recorded task completion, visible hesitation, navigation errors, recovery success, subjective confidence, and comments about trust in the AI output. For heuristic findings, severity was classified as Critical, Serious, Moderate, or Minor according to impact on task completion and frequency of occurrence.

3.2 Evaluation Results

The evaluation found that the overall concept was understandable and relevant to the target users, especially because participants recognized the value of role-specific questions and structured feedback. However, the prototype still created friction in three areas: insufficient control over interview context, weak protection of in-progress work, and high cognitive load when too many AI coaching elements appeared at once.

Dimension	Score	Key Finding	Design Implication
Consistency	3 / 5	AI feedback cards used different layouts in Practice and Review, making users re-learn similar information patterns.	Standardize the feedback card structure across workflows.
Status Feedback	3 / 5	The Practice setup wizard did not clearly show progress or the current step.	Add a step indicator and visible state changes during generation.
User Control	2 / 5	Users could not edit the job description or resume context after question generation without restarting.	Allow context editing and regeneration without discarding the whole session.
Error Prevention	2 / 5	Answer drafts could be lost when navigating away from an in-progress response.	Introduce autosave, warning dialogs, and recoverable drafts.
Simplicity	3 / 5	Five simultaneous coaching panels created overload during answer revision.	Prioritize one primary coaching view and reveal deeper analysis progressively.
General Usability	3 / 5	The 1024px layout was crowded and history acted more like storage than reflection support.	Improve responsive layout and redesign history as an active comparison tool.

Severity Classification

ID	Severity	Problem	Evidence	Iteration Response
C-01	Critical	No way to edit JD or resume context after question generation.	Users expected to correct context when a generated question felt irrelevant.	Context editing and regeneration controls were added.
C-02	Critical	Navigating away from an in-progress answer discarded the draft silently.	Participants hesitated after losing text and became less willing to explore.	Autosave, leave warning, and draft restore behavior were specified.

S-01	Serious	The Practice setup wizard lacked a clear progress indicator.	First-time users were unsure how many setup steps remained.	A three-step progress indicator was added.
S-02	Serious	Too many coaching panels appeared at the same time.	Participants scanned repeatedly and asked which panel mattered most.	Real-time coaching was reorganized around STAR coverage and priority tips.
M-01	Moderate	Feedback cards used inconsistent layouts across Practice and Review.	Users slowed down when switching between workflows.	A unified feedback-card template was introduced.
M-02	Moderate	AI confidence cues were visible but not always explained.	Participants wanted to know why confidence was low.	Confidence explanations were paired with observable evidence.
M-03	Moderate	Question lists lacked filtering for weak areas or favorites.	Users wanted to return quickly to difficult questions.	Question management tools were added.
N-01	Minor	Some labels used product vocabulary that was not immediately clear.	Participants paused at terms such as 'gap' and 'coverage'.	Labels were simplified and paired with short contextual descriptions.
N-02	Minor	The layout became dense around 1024px width.	Side panels compressed answer content.	Spacing and panel priority were adjusted for medium screens.
N-03	Minor	History did not show enough comparison between attempts.	Users wanted to see whether they had improved.	History was redesigned into a comparison studio.

Cognitive Walkthrough Findings

The walkthrough showed that users could identify the primary action in most screens, but breakdowns occurred when the next step depended on hidden system state. In the Practice workflow, users understood that they needed to provide job context, but they could not predict whether changing that context later would preserve previous answers. In the Review workflow, users understood the purpose of transcript verification, but the prototype needed clearer feedback after edits were saved. These findings strengthened the need for persistent state, visible progress, and explicit confirmation messages.

Think-Aloud Findings

Observation	Participant Evidence	Interpretation
Users valued personalization.	All 3 participants responded positively when questions referenced resume or job-description details.	Personal context is a key trust signal and should remain visible throughout the workflow.
Users wanted guidance while answering, not only after submission.	2 participants asked whether their answer was 'on the right track' before submitting.	Real-time coaching can reduce uncertainty and unnecessary re-submissions.
Users were cautious	Participants accepted scores more	Scores should be treated as explainable

about AI scoring.	readily when reasoning and confidence were shown together.	suggestions rather than final judgments.
Users wanted evidence of progress.	All 3 participants looked for a way to compare previous and current attempts.	History should support reflection, comparison, and targeted re-practice.

3.3 Iteration Evidence

Three design iterations were completed between prototype v1 and v2. Each iteration directly addressed a usability issue found in Section 3.2 and was checked against the product goal of making InterviewAI transparent, controllable, and useful for repeated preparation.

Iteration	Evaluation Trigger	Design Change	Expected Improvement
1. Question Management	S-02 and M-03: users felt overloaded and could not quickly return to weak questions.	Added filter, sort, favorite, and mark-as-weak controls to the question page. Questions can now be grouped by type, difficulty, status, and relevance to the target role.	Users can focus on high-priority questions instead of scanning the full generated list every time.
2. Real-Time Coaching	S-02 and think-aloud evidence: users wanted to know whether an answer was strong before submitting.	Shifted AI support from post-hoc judging to in-process coaching. Added STAR coverage detection, priority coaching tips, confidence explanations, and quick improvement inserts.	Users receive timely guidance while drafting, which reduces uncertainty and repeated trial-and-error submissions.
3. History Comparison Studio	General usability score of 3 and participant demand for progress evidence.	Changed History from a passive archive into a side-by-side comparison studio with score trend, answer differences, recurring gaps, and 'Practice Similar Gap' entry points.	Users can reflect on improvement over time and convert past weaknesses into new practice tasks.

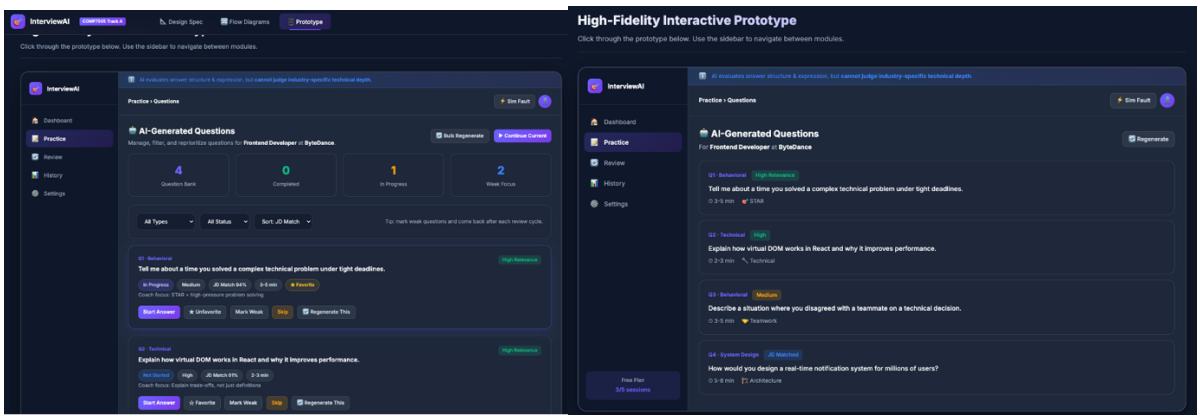

Figure D3-1: IT-1 Question Management — Before (static cards, no filtering) vs. After (filters, favorites, weak-area tags, quick actions)

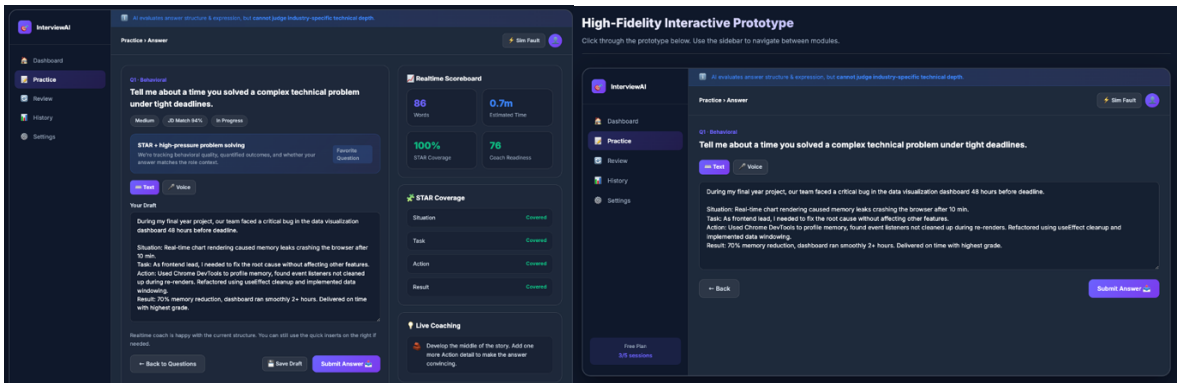

Figure D3-2: IT-2 Real-time Coaching — Before (single-column, feedback only after submission) vs. After (two-column layout with live STAR coaching panel)

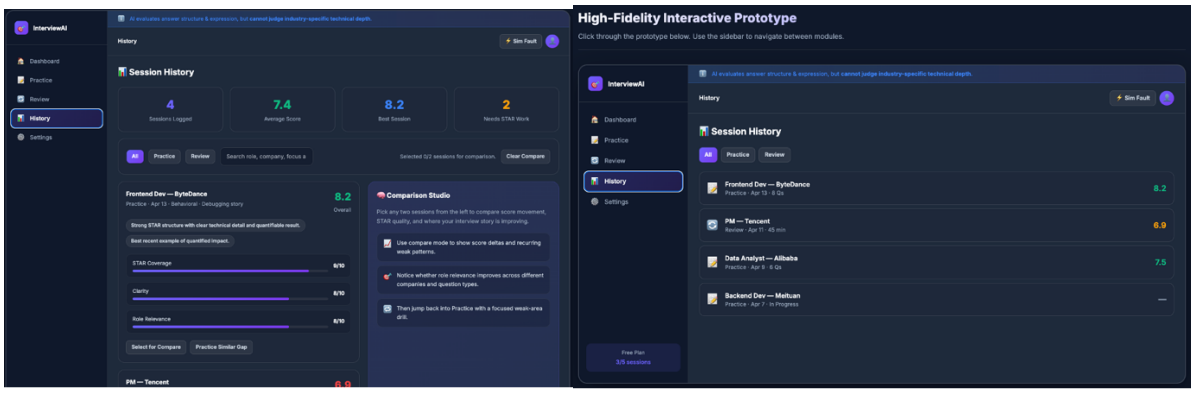

Figure D3-3: IT-3 History Comparison — Before (simple record list, no comparison) vs. After (side-by-side comparison studio with progress tracking)

Iteration 2 Measurement

Metric	Before v1	After v2	Interpretation
Average re-submissions per question	2.3	1.1	Real-time coaching reduced repeated blind revisions.
Participants reporting 'I do not know if my answer is good'	3 /	0 / 3	STAR coverage and priority tips improved confidence during drafting.
Participants who used the STAR panel	Not available	3 /	The new coaching panel was discoverable and useful.
Participants who could identify their weakest answer area	1 / 3	3 /	The history comparison view made progress and gaps more visible.

Before-and-After Summary

- Before iteration, the prototype provided many AI outputs but did not always make the user's next action obvious. After iteration, each major output is paired with an action such as edit context, improve answer, save draft, compare attempts, or practice a similar gap.
- Before iteration, the system behaved mainly as an evaluator after the user submitted an answer. After iteration, the system also supports the answer-construction process through lightweight coaching and STAR coverage checks.
- Before iteration, history mainly stored past sessions. After iteration, history became an active reflection workspace that helps users understand patterns across attempts.

Overall, Deliverable 3 demonstrates a complete evaluation-to-iteration cycle: findings were collected through complementary methods, prioritized by severity, converted into design changes, and checked against the central design principles of transparency, user control, and graceful degradation.

Deliverable 4: Design Principles & Product Roadmap

This deliverable presents three reusable interaction design principles derived from our project experience, each supported by concrete design examples from InterviewAI. It also outlines a phased product development roadmap for future expansion.

4.1 Reusable Design Principles

Principle 1: Transparency of AI Reasoning

Principle: Every AI output must include a reasoning explanation and a confidence indicator. Users should be able to judge the relevance and reliability of AI suggestions rather than blindly trusting them.

Rationale

This principle addresses the fundamental trust gap identified in the project research. Users were concerned that AI feedback could feel unreliable, generic, or insufficiently grounded in their job description and resume. In the survey evidence used by the team, **6 out of 9 respondents reported that AI feedback felt unreliable**. For InterviewAI, this means that AI cannot be presented as a black-box judge. It must behave as an inspectable coach.

Reusable application

For future modules, every important AI judgement should follow a four-part structure:

1. AI judgement — score, suggestion, category, or warning;

2. Reasoning — concise explanation based on observable evidence;

3. Confidence — High / Medium / Low or another uncertainty signal;

4. User verification — limitation statement or invitation for human judgement.

Principle 2: User Control Over AI Outputs

Principle: Users must always have final control over AI-generated evaluations. They should be able to flag inaccuracies, override scores, undo actions, and discard flawed evaluations. AI should suggest, not dictate.

Rationale

This principle directly responds to the frustration captured in the persona and evaluation findings: users lose trust when AI appears to ignore resume or job-description context. One representative user concern was: “Sometimes the AI clearly didn’t read my resume... That breaks trust immediately.”

The usability evaluation also found several control-related issues:

- users could not edit JD or resume context after question generation without restarting;

- answer drafts could be lost when navigating away;

- users needed more ways to challenge, revise, or recover from AI-generated content.

Therefore, user control is not a secondary usability detail. It is central to trust and task completion.

Reusable application

Any future AI-generated output should be designed with the following checklist:

- Can the user challenge this output?

- Can the user revise or regenerate it?

- Can the user undo or discard the result?

- Can the user preserve work before leaving the page?

- Can the user return to an earlier state without losing all progress?

Principle 3: Graceful Degradation

Principle: System failures must provide clear degradation paths with no dead ends. When AI or recognition capabilities fail, the system should offer meaningful fallbacks such as text input for voice failure, a generic question bank for generation errors, or manual Q&A entry for extraction failure.

Rationale

Interview preparation often happens under time pressure. A technical failure should not stop the user’s preparation session. The Track A requirements also emphasise capability boundaries, failure handling, and visible limitation communication. For InterviewAI, graceful degradation is essential because several product functions depend on uncertain AI or

recognition capabilities, including question generation, voice recognition, and audio extraction.

Reusable application

For future features, every AI-dependent workflow should define:

- the likely failure mode;

- the user-facing message;

- the fallback action;

- the state that should be preserved;

- the recovery path back into the main workflow.

4.2 Product Development Roadmap

Our product expansion plan is organised into three phases. Each phase builds on the previous one while maintaining the three core design principles: transparent AI reasoning, user control over AI outputs, and graceful degradation.

Phase	Focus	Description
Phase 1	MVP Enhancement	Real-time voice analysis for spoken answers; multi-language support (English + Chinese); collaborative practice rooms for peer mock interviews
Phase 2	Intelligent Adaptation	AI-driven difficulty scaling based on performance history; personalized learning paths; industry-specific question banks (finance, tech, consulting); integration with calendar apps for scheduling
Phase 3	Ecosystem Expansion	University career center partnerships; employer-branded practice modules; alumni mentor matching; analytics dashboard for career services; mobile app for on-the-go practice

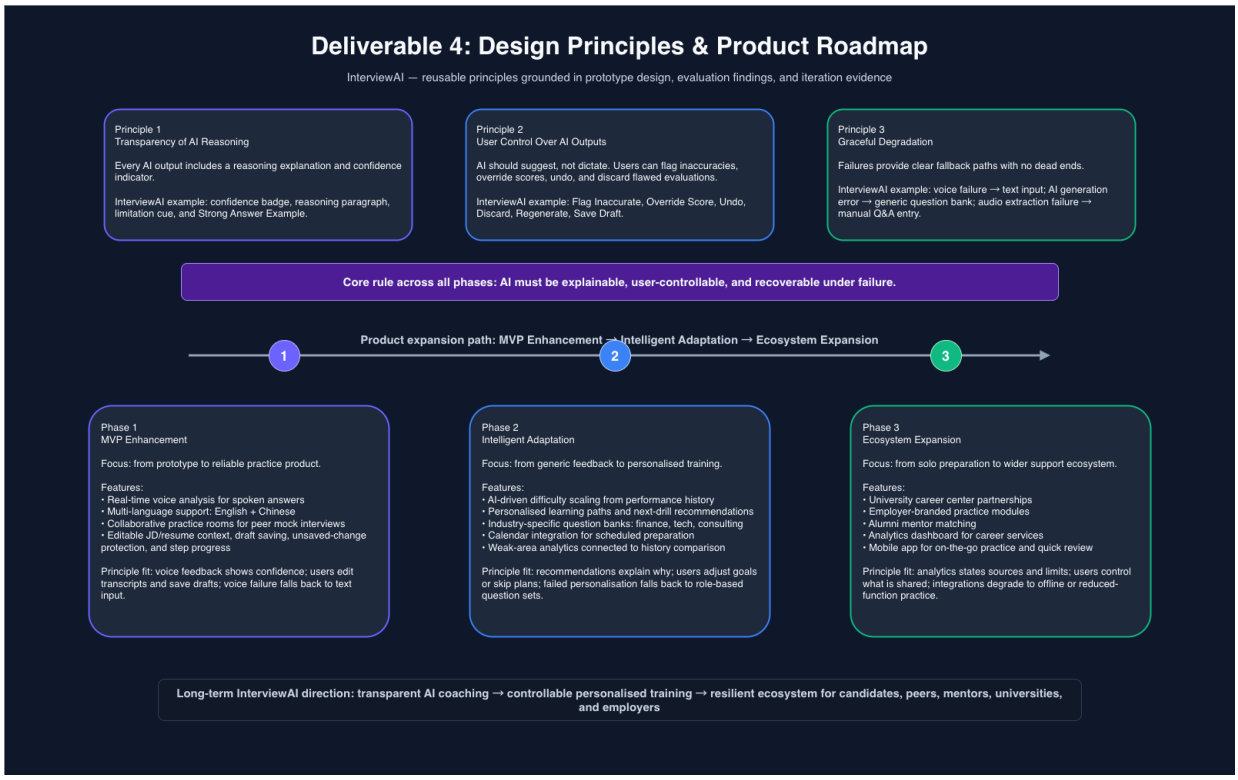

Figure D4-1: Roadmap